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Lab 01:

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178 --Lab 01:
179 -- Tìm một rental để test
180 SELECT rental_id, customer_id FROM rental LIMIT 1;
181 --Step 2
182 BEGIN;
183
184 UPDATE rental
185 SET return_date = NOW()
186 WHERE rental_id = 1;
187
188 INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
189 VALUES (1, 1, 1, 5.99, NOW());
190
191 COMMIT;
192 --Step 3:
```

Data Output Messages Notifications

	rental_id [PK] integer	customer_id smallint
1	2	459

```
181 --Step 2
182 BEGIN;
183
184 UPDATE rental
185 SET return_date = NOW()
186 WHERE rental_id = 1;
187
188 INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
189 VALUES (1, 1, 1, 5.99, NOW());
190
191 COMMIT;
192 --Step 3:
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 109 msec.

	rental_id [PK] integer	return_date timestamp without time zone
1	1	2026-09-18 14:55:36.326746

```

192 --Step 3:
193 BEGIN;
194
195     UPDATE rental
196     SET return_date = NOW()
197     WHERE rental_id = 1;
198
199     -- Cố tình gây lỗi: staff_id không tồn tại (vi phạm Foreign Key)
200     INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
201     VALUES (1, 9999, 1, 5.99, NOW());
202
203 COMMIT;
204 SELECT rental_id, return_date FROM rental WHERE rental_id = 1;
205
206 --Ex 1:

```

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ERROR: insert or update on table "payment" violates foreign key constraint "payment_staff_id_fkey"
Key (staff_id)=(9999) is not present in table "staff".

SQL state: 23503

Detail: Key (staff_id)=(9999) is not present in table "staff".

Ex 1:

```

206 --Ex 1:
207 SELECT return_date FROM rental WHERE rental_id=2;
208 BEGIN;
209 --Step 1: Mark as returned
210 UPDATE rental SET return_date = NOW() WHERE rental_id = 2;
211
212 --Step 2: Intentional Error (Invalid payment_id or Column)
213 INSERT INTO payment (customer_id, staff_id, rental_id, amount,
214 payment_date)
215 VALUES (459, 1, 2, 99.99, 'WRONG_DATE_FORMAT');
216
217 COMMIT;
218
219 --Ex 2:
220 BEGIN;
221

```

Data Output Messages Notifications

Icons: +, document, dropdown, clipboard, dropdown, trash, database, download, line graph, SQL

	return_date timestamp without time zone
1	2005-05-28 19:40:33

```

206  --Ex 1:
207  SELECT return_date FROM rental WHERE rental_id=2;
208  BEGIN;
209  --Step 1: Mark as returned
210  UPDATE rental SET return_date = NOW() WHERE rental_id = 2;
211
212  --Step 2: Intentional Error (Invalid payment_id or Column)
213  INSERT INTO payment (customer_id, staff_id, rental_id, amount,
214  payment_date)
215  VALUES (459, 1, 2, 99.99, 'WRONG_DATE_FORMAT');
216
217  COMMIT;
218
219  --Ex 2:

```

Data Output Messages Notifications

```

ERROR:  invalid input syntax for type timestamp: "WRONG_DATE_FORMAT"
LINE 8: VALUES (459, 1, 2, 99.99, 'WRONG_DATE_FORMAT');
                                   ^

```

```

SQL state: 22007
Character: 257

```

Q: Is the return_date updated? Why not?

The return_date was because the entire transaction failed and was rolled back due to the error in the subsequent INSERT statement.

Ex 2:

```
SAVEPOINT
```

```
Query returned successfully in 66 msec.
```

```

220 BEGIN;
221 -- Bước 1: Đăng ký nhân viên mới (tạm thời gán store_id đúng để insert thành công trước)
222 INSERT INTO staff (first_name, last_name, address_id, email, store_id, active, username, password)
223 VALUES ('Nguyen', 'An', 1, 'nguyen.an@example.com', 1, TRUE, 'nguyenan', 'password123');
224
225 -- Bước 2: Đặt SAVEPOINT trước khi thử gán store_id
226 SAVEPOINT before_store_assignment;
227
228 -- Bước 3: Thử gán nhân viên vào store_id không tồn tại -> sẽ lỗi vì vi phạm Foreign Key
229 UPDATE staff
230 SET store_id = 999
231 WHERE staff_id = (SELECT staff_id FROM staff WHERE username = 'nguyenan');
232
233 -- ERROR: insert or update on table "staff" violates foreign key constraint
234 -- DETAIL: Key (store_id)=(999) is not present in table "store".
235
236 -- Bước 4: Vì lệnh trên lỗi, ROLLBACK về đúng SAVEPOINT (không hủy toàn bộ transaction)
237 ROLLBACK TO SAVEPOINT before_store_assignment;
238
239 -- Gán lại đúng vào Store 1
240 UPDATE staff
241 SET store_id = 1
242 WHERE username = 'nguyenan';
243
244 -- Bước 5: COMMIT toàn bộ transaction
245 COMMIT;

```

Data Output Messages Notifications

COMMIT

Query returned successfully in 66 msec.

246 select * from staff;

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1									
	staff_id [PK] Integer	first_name character varying (45)	last_name character varying (45)	address_id smallint	email character varying (50)	store_id smallint	active boolean	username character varying (16)	password character varying (40)
1	1	Mike	Hillyer	3	Mike.Hillyer@sakilastaff.com	1	true	Mike	8cb2237d0679ca88db6464eac6
2	2	Jon	Stephens	4	Jon.Stephens@sakilastaff.c...	2	true	Jon	8cb2237d0679ca88db6464eac6
3	3	Nguyen	An	1	nguyen.an@example.com	1	true	nguyenan	password123

Lab 1 update:

```

250 -- Tìm một rental để test
251 SELECT rental_id, customer_id FROM rental LIMIT 1;
252 BEGIN;
253
254 -- Bước A: Update trạng thái rental thành 'returned'
255 UPDATE rental
256 SET return_date = NOW()
257 WHERE rental_id = 1;
258
259 -- Đặt SAVEPOINT trước khi thử insert payment
260 SAVEPOINT before_damage_fee;
261
262 -- Bước B: Cố tình "phá" bước này để test Atomicity - staff_id không tồn tại
263 INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
264 VALUES (1, 9999, 1, 5.99, NOW());
265
266 -- ERROR: insert or update on table "payment" violates foreign key constraint
267 -- DETAIL: Key (staff_id)=(9999) is not present in table "staff".
268 -- Quay lại điểm mốc trước khi insert lỗi (giữ nguyên UPDATE rental ở Bước A)

```

Data Output Messages Notifications

SAVEPOINT

Query returned successfully in 62 msec.

```

263 INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
264 VALUES (1, 9999, 1, 5.99, NOW());
265
266 -- ERROR: insert or update on table "payment" violates foreign key constraint
267 -- DETAIL: Key (staff_id)=(9999) is not present in table "staff".
268 -- Quay lại điểm mốc trước khi insert lỗi (giữ nguyên UPDATE rental ở Bước A)
269 ROLLBACK TO SAVEPOINT before_damage_fee;
270
271 -- Thử lại insert payment với staff_id hợp lệ
272 INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
273 VALUES (1, 1, 1, 5.99, NOW());
274
275 COMMIT;
276 SELECT rental_id, return_date FROM rental WHERE rental_id = 1;
277 SELECT * FROM payment WHERE rental_id = 1 ORDER BY payment_date DESC LIMIT 1;

```

Data Output Messages Notifications

ERROR: insert or update on table "payment" violates foreign key constraint "payment_staff_id_fkey"
Key (staff_id)=(9999) is not present in table "staff".

SQL state: 23503

Detail: Key (staff_id)=(9999) is not present in table "staff".

ROLLBACK

Query returned successfully in 47 msec.

```

272 INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
273 VALUES (1, 1, 1, 5.99, NOW());
274
275 COMMIT;
276 SELECT rental_id, return_date FROM rental WHERE rental_id = 1;
277 SELECT * FROM payment WHERE rental_id = 1 ORDER BY payment_date DESC LIMIT 1;

```

Data Output Messages Notifications

INSERT 0 1

Query returned successfully in 60 msec.

```

275 COMMIT;
276 SELECT rental_id, return_date FROM rental WHERE rental_id = 1;
277 SELECT * FROM payment WHERE rental_id = 1 ORDER BY payment_date DESC LIMIT 1;

```

Data Output Messages Notifications

Showing rows: 1 to 1

	rental_id [PK] integer	return_date timestamp without time zone
1	1	2026-09-18 15:33:55.99946

Showing rows: 1 to 1

	payment_id [PK] integer	customer_id smallint	staff_id smallint	rental_id integer	amount numeric (5,2)	payment_date timestamp without time zone
1	32102	1	1	1	5.99	2026-09-18 15:33:55.99946